Multi-Domain Inconsistency Framework for Image Forgery Detection

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***Abstract*—Image forgery creates digital mistrust in this era of widespread AI-generated content, from GAN-based deepfakes to diffusion model outputs that almost remarkably look exactly like real photographs. Conventional detectors may falter when brought face-to-face with the latest synthetic content, with cases of compression or local manipulation or even full-frame fakes. We propose a Multi-Domain Inconsistency Fusion (MDIF) framework, a lightweight hybrid network that simultaneously processes RGB spatial features via CNN, DCT/DFT frequency spectra for generation artifacts, and MiDaS-derived depth maps for geometric inconsistencies, capable of detecting AI images. Using datasets including GenImage, CIFAKE, AutoSplice, and CocoGlide, we hypothesise that the model could achieve a high accuracy with great efficiency.**

***Index Terms*—Image Forgery Detection, Multi-Domain Inconsistency, Hybrid Network, AI Images**

I. Introduction

Recent developments in the world of Artificial Intelligence and its ability to generate almost undetectable images have taken over the world by storm, with every field relying on technology to make content easily and within minutes. When it comes to image generation, however, AI tools today can fabricate convincing forgeries for disinformation, "deepfakes", and fraud just as easily. Hence, the question, where do we draw the line on synthetic content? Traditionally, digital forensics has relied on detecting copy-move duplication, splicing composites and object removal, left spatial inconsistencies, using CNN architectures, but now diffusion models like Stable Diffusion and DALL·E produce globally synthetic images indistinguishable to the human eye which poses a problem.

On March 23, 2023, certain images were leaked of American President Donald Trump and Russian President Vladimir Putin getting tackled by police officers [1]. While the images become widely sensational, they were AI generated and such situations invoke mistrust in people unable to identify what is real and what is not. Along with image generation, AI-powered inpainting, which is the selective modification of authentic photographs, has become accessible to all and hugely popular, with Adobe Photoshop's Generative Fill and Google's Gemini and many other such tools. The consequence of undetected image forgeries is a matter of serious concern, where political misinformation, manipulated imagery in civil disputes and even faked identity documents can cause heavy harm and damage to reputation.

Existing detection models, even state-of-the-art systems like FatFormer, GRAMNet, and DiffCoR, face two critical limitations: they generalize poorly to generator models not seen during training and several require significant computational resources unavailable in resource-constrained environments. A computationally efficient, generalizable, and inpainting-aware detector is therefore a pressing requirement.

AI-generated and AI-manipulated (inpainted) images that are produced by diffusion-based generative models must be reliably detected by a system with good accuracy and domain generalization capability. The system must be able to classify images into three broad categories: not AI-generated, AI-generated, or AI-manipulated. The detection must be robust to image post-processing operations like compression or resizing, and able to identify AI-generated images across different generation architectures.

Current AI image generators are too powerful, and people cannot distinguish between what is real and what has been generated. There is a need to make detection automatic and algorithm-driven in this scenario. Failure to detect such forgeries will have bearing consequences in the fields of news and journalism, legal junctures, fraud and crime prevention, social media integrity and many other relevant spheres of life. The gap in the detection of inpainted images is especially worrisome as bad actors can manipulate select regions of an image to their liking while also being able to pass them as credible against traditional verification algorithms.

II. Literature Review

Modern day misinformation need not be only altered images. It can also be the captions attached to unaltered image. Sagar et al. [2] aims to verify the efficacy of the use of IMD techniques in the context of "Multimodal Misinformation". This is achieved in a two-stage process starting with a MCTS multimodal LLM supported by web search and similar RAG tools, and multiple rounds of "Multi-Agent Debate" between a sceptic agent and a supporter agent. This new architecture is tested against five IMD techniques, namely, NPR [3], BNext-M [4], GRAMNet [5], ProDet [6], and FatFormer [7]. These are then benchmarked on the MMFakeBench and the DGM4 datasets.

In Sun et al. [8], the authors address limitations of using multimodal large language models (M-LLMs) to perform image manipulation detection (IMD) tasks by proposing "ForgerySleuth", a tool that aims to provide a much-detailed explanation of detected clues and segments the regions that have been tampered. They have also constructed a more diverse dataset, aptly titled "ForgeryAnalysis Dataset" from existing datasets including CASIA [9], AutoSplice [10], DEFACTO [11], and MIML [12].

While the field of image synthesis with the use of artificial intelligence has been rapidly evolving thanks to the diffusion-based generative models, identification of such content becomes crucial in image forensics. To address the increased existence of GAN produced images, Tran et al. [13] propose DiffCoR. This detection method uses Stable Diffusion Processing (SDP) module, which uses a pre-trained model to reconstruct images using reverse diffusion and Image Representation Learning (IRL) that applies Self-Supervised Learning (SSL) with Latency Consistency Loss (LCL). Additionally, Discrete Fourier Transform (DFT) allows a frequency domain analysis for enhancing the detection of any manipulation. The dataset introduced for this research is "ForensicsImage", which constitutes of both real and AI generated images from LSUN-Bedroom, CelebA-HQ and CelebDFv2, among other diffusion models like LDM [14], DALL-E [15], Imagen [16], and Midjourney.

In contrast to the deep learning framework, DiffCoR suggests, Bammey [17] brings forward the idea of a computationally lean, interpretable detector optimized for photorealistic diffusion outputs from models like Stable Diffusion 1.3-XL [14], Midjourney v5, DALL·E 2/3 [15], Glide, and Adobe Firefly, using a cross-difference high-pass filter to highlight frequency artifacts at periods 2, 4, and 8 in the Fourier transform of a residual image, allowing a clear distinction of a real image from a fake one. In addition, it extracts 135 peak magnitudes per image as features for a lightweight HBGB classifier and handles JPEG compression through per-quality-factor models. The research used a self-constructed dataset from which the synthetic images were generated from text prompts, while the real images were borrowed from the Raise-1k dataset, a subset of the Raise [18] dataset. The architecture brings a good generalization to the table but focuses only on diffusion models.

While methods such as those proposed by Bammey [17] work effectively on high-frequency artifacts, these details are at risk of being lost during post-processing like image compression. For detection to work even after such alterations and to expand detection beyond just diffusion models, Manisha et al. [19] propose an approach based on source camera fingerprinting. Fingerprints (artifacts and other information) left on photographs by physical cameras and AI models are very different; by focusing on mid-frequency details, this difference can be discerned, even after compression. Since each AI model will have its unique fingerprint, this methodology also makes it possible to identify the exact model that has generated an image. A ResNet101 model is trained on QUFVD images resized to 224x224 pixels. The resizing strips high frequency data and forces the model to learn mid-frequency fingerprints. The trained ResNet101 outputs a 2048-dimensional fingerprint, which is then fed into the SVM, for a set of real (FFHQ) and fake (DH3) images, to build the classification model. On the DF3 and CNN-aug datasets, the model has an average AUC of 0.994 and showcases high resilience to image alterations. However, the reliance on fingerprints compromises the accuracy of the model when there are localized image changes such as face blending.

Building upon the need to capture localized artifacts that methods like Manisha et al. [19] struggle with while still making use of frequency-domain analysis, Yan et al. [20] propose a Dual Frequency Branch Framework combined with Reconstructed Sliding Windows Attention (RSWAttention). Unlike earlier single-frequency approaches that could lose data during compression or resizing, this architecture extracts many perspective features by fusing the Discrete Wavelet Transform (DWT) for local high-frequency and low-frequency textures, with the phase component of the Fast Fourier Transform (FFT). To isolate and find subtle tampering, RSWAttention restricts feature extraction to the local window, which forces the model to evaluate how the neighboring pixels are interdependent. It was evaluated against 65 distinct GAN and diffusion models, and showed good generalization, better than baselines such as FatFormer by Liu et al. [7]. Despite the local attention mechanism, the framework has vulnerabilities with source fingerprinting while detecting highly localized face-swapping alterations.

III. Dataset Description

This section describes the datasets used in this project, including their sources, characteristics, and how they are combined to train the models in the proposed framework.

A. Raw Datasets

Table I describes some relevant statistics of the combined datasets used to train the models in the proposed framework. The dataset contains images which pertain to three categories: authentic images, fully AI-generated images and AI-manipulated (inpainted) images, all collected from multiple publicly available benchmark datasets.

Table I
DATASET CHARACTERISTICS

	Column Name	Description	Train Set Count	Test Set Count
1	CIFAKE [21]	Authentic and Generated Images	100,000	20,000
2	Unbiased Tiny GenImage [22]	Authentic and Generated Images	8,662	2,165
3	AutoSplice [10]	Authentic and Inpainted Images	5,519	376
3	CocoGlide [23]	Authentic and Inpainted Images	819	205
3	SAGI [24]	Authentic and Inpainted Images	224,484	12,688

1) CIFAKE: The CIFAKE dataset [21] constitutes two classes:

- REAL - the authentic photographs
- FAKE - AI-generated synthetic images

The primary use of this dataset was to improve the mode's ability to differentiate between images which are real from images which are entirely synthetic. CIFAKE produces strong images from global image synthesis artifacts produced by GAN and diffusion-based generators.

2) Unbiased Tiny GenImage: From the GenImage dataset [22], the following subsets were used:

- Nature - authentic images
- Midjourney - AI-generated images
- Glide - AI-generated images

The purpose of this dataset was primarily to improve cross-generator generalization. Since the generated images were from different diffusion architecture origins, the model was able to learn the spectral and spatial inconsistencies from the images rather than memorizing artifacts from a single model.

3) AutoSplice: The AutoSplice dataset [10] was source for:

- Authentic - real images
- Forged_JPEG90 - AI-manipulated images

The focus of this particular dataset was on localized image manipulation and splicing operations that helped the model learn discontinuities introduced during selective image edition and region replacement.

4) CocoGlide: The CocoGlide dataset [23] covered:

- Real - authentic images
- Fake - inpainted images

The AI-manipulated samples are generated using GLIDE's inpainting techniques, which was essential to the framework in teaching the model subtle boundary inconsistencies and abnormal texture transitions, created in the AI-assisted editing process.

5) SAGI: The SAGI dataset [24] contains subsets such as:

- original - authentic images
- brushnet - inpainted images
- controlnet - inpainted images
- removeanything - inpainted images

The RAISE split of the dataset was the only category used during training. SAGI introduced multiple modern inpainting approaches that contributed to the model's ability to generalize across different pipelines and editing strategies.

B. Processed Data

Once the datasets were acquired, preprocessing was performed before training the framework. The preprocessing pipeline included:

1) Image resizing and normalization
2) Extraction of spectral features using Discrete Cosine Transform (DCT)
3) Generation of monocular depth maps using MiDaS
4) Feature vector construction for hybrid fusion

The processed dataset was stored in separate train and test directories. Each sample contains a resized image file (.jpg), along with its corresponding extracted feature files (.npy).

Most datasets were split using an 80-20 train-test ratio. The training split was further divided into 80% training and 20% validation during model training.

Table II
LABEL DISTRIBUTION IN THE COMBINED DATASET

	Label	Numeric Label	Train Set Count	Test Set Count
1	Authentic	0	122,475	15,226
2	AI Generated	1	54,797	11,797
3	Inpainted	2	38,083	2,494

The files follow a consistent naming scheme: <label>_<original_file_name>, where 0 indicates authentic images, 1 stands for AI-generated images and 2 is the label for inpainted images as shown in table II. This preprocessing strategy essentially ensures efficiency in feature loading during model training and subsequently reduces repeated computation overhead.

IV. METHODOLOGY

Figure 1 shows the brief architecture of the proposed Multi-Domain Inconsistency Fusion (MDIF) framework. The image will take three paths, combining spatial, spectral and geometric inconsistencies to attain the lightweight hybrid detection pipeline.

The input image simultaneously goes through three independent streams:

1) Spatial Feature Stream
2) Spectral Feature Stream
3) Depth Consistency Stream

From which the extracted features are fused into a concatenated representation, classified using a lightweight multilayer perceptron classifier.

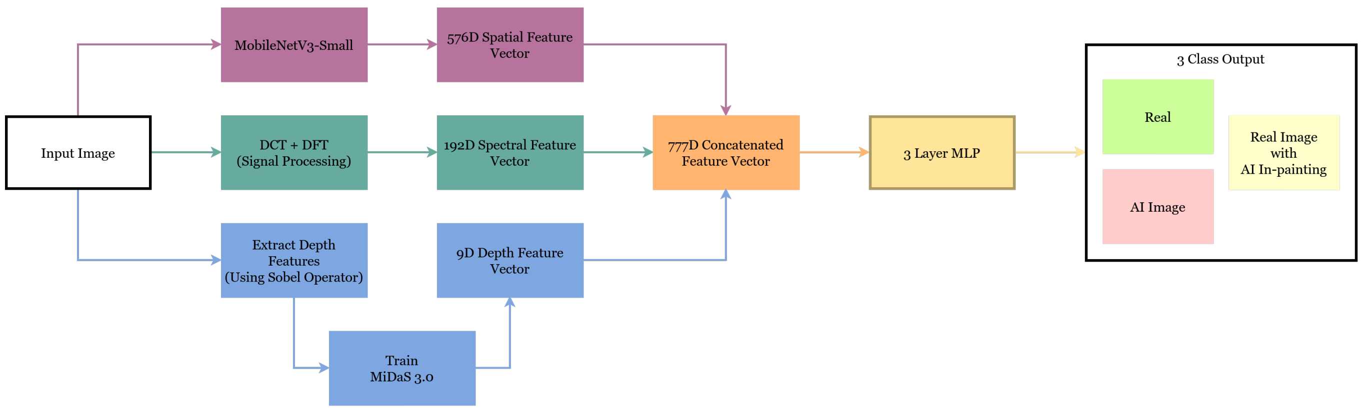

Figure 1. Pipeline of the Hybrid Framework explained in detail in the sections IV-A, IV-B and IV-C.

A. Stream A: Spatial Features

The first path is to a Spatial RGB CNN, a MobileNetV3Small model pretrained on ImageNet. The network extracts 576-dimensional spatial feature vectors representing texture inconsistencies, abnormal semantic structures, blending artifacts, overly-smoothed regions and unrealistic object details. MobileNetV3Small's lightweight architecture and low computational overhead makes it capable for strong feature extraction and suitable for complex systems, often with resource constraints, adding to its idealness for this framework.

B. Stream B: Spectral Features

DCT (Discrete Cosine Transform) helps identify the smoothing artifacts of the diffusion models along with periodic spectral irregularities. The frequency of discontinuities across manipulated regions and the loss of natural camera frequency distributions adds onto DCT's 192-dimensional spectral feature vector. This frequency-domain analysis is key in detecting traces in low and mid frequency components in images which AI generation may leave behind, even when visual inspections are unable to identify them.

C. Stream C: Depth Features

The final path is through the MiDaS model to generate a monocular depth map. Gradient magnitude maps are computed from the RGB image and the depth map using a Sobel-Feldman operator as seen in equation 3. A consistency map is created by computing the correlation scores from sliding a 16x16 window across the image. This is then used to create a 9-dimensional discrepancy vector that captures the logically implausible depth-shading relationships in images that diffusion models generate.

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} \tag{1}$$

$$G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} \tag{2}$$

$$G = \sqrt{(G_x)^2 + (G_y)^2} \tag{3}$$

AI-generated and inpainted images fail to maintain the realistic depth and shading consistency in most cases, making this stream effective in manipulation detection.

D. Hybrid Classifier

These feature vectors are then concatenated to a 777-dimensional feature vector which is to a simple 3-Layer MLP (Multilayer Perceptron) with ReLU activations and dropouts. The final output will be a three-class probability distribution over "Real", "Fully AI Generated", and "Real with AI Inpainting".

V. Implementation Details

A. Experimental Setup

The project is developed in Python and utilizes libraries mentioned in table III and implemented on a machine with the specifications shown in the table IV.

Table III
SOFTWARE DEPENDENCIES

	Libraries	Versions	Description
1	PyTorch	2.11.0+cu130	Machine Learning Framework
2	Torchvision	0.26.0+cu130	Computer Vision Library
3	OpenCV	4.13.0	Computer Vision Library
4	Pillow	12.1.1	Image Processing Library
5	Scikit-learn	1.8.0	Machine Learning Library
6	NumPy	2.4.3	Numerical Computing Library
7	SciPy	1.17.1	Scientific Computing Library
7	Matplotlib	3.9.1	Data Visualization Library
7	Seaborn	0.13.2	Data Visualization Library

B. Evaluation Metrics

The framework is evaluated on accuracy, area under the ROC curve, precision, recall, and F1-score metrics. This concept has achieved an accuracy of 92% and an AUC of 0.98 on the test set. The full detailed results of the metrics will be discussed in section VI.

Table IV
HARDWARE SPECIFICATIONS

	Hardware/Software	Specifications
1	Operating System	Windows 10/11
2	Programming Language	Python 3.11
3	Memory	32GB
4	CPU	AMD Ryzen 9 7940HX 2.4 GHz
5	GPU	NVIDIA GeForce RTX 4070 Laptop 8GB
6	Environment	Miniconda

1) Accuracy:

$$\text{Accuracy} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \tag{4}$$

Where, T_P stands for True Positives, T_N for True Negatives, F_P for False Positives and F_N for False Negatives. Accuracy, as defined in Equation 4, is the ratio of correctly classified instances, i.e. the sum of true positives (T_P) and true negatives (T_N), to the total number of instances.

2) Area Under the ROC Curve: The Receiver Operating Characteristic (ROC) Curve is a visualization of the performance of a binary classifier across different thresholds.The Area Under the ROC Curve (AUC) is a metric that summarizes the overall performance of the classifier. An AUC of 1.0 shows perfect classification, while an AUC of 0.5 shows random guessing. The AUC is calculated by integrating the area under the ROC curve, which plots the True Positive Rate ($T_P R$) against the False Positive Rate ($F_P R$) at various threshold settings. A higher AUC value indicates better model performance in distinguishing between the positive and negative classes.

3) Precision: Precision is the ratio of true positives to the sum of true and false positives, measuring the accuracy of the positive predictions made by the model, as shown in the equation

$$\text{Precision} = \frac{T_P}{T_P + F_P} \tag{5}$$

Where, T_P stands for True Positives and F_P for False Positives. A high precision is an indicator for a model that makes fewer false positive errors, meaning that when it predicts a positive class, it is likely to be correct.

4) Recall: Recall is that machine learning metric, evaluating a model's ability to find all relevant cases in a dataset, mathematically calculated as the ratio of true positives to the total number of actual positives, which is the sum of true positives and false negatives in the predictions. Recall is calculated using equation 6

$$\text{Recall} = \frac{T_P}{T_P + F_N} \tag{6}$$

Where, T_P stands for True Positives and F_N for False Negatives.

5) F1-Score: That evaluation metric in machine learning defined as the harmonic mean of precision and recall is the F1 Score, combining both into a balanced score ranging from 0 to 1; 0 being the worst and 1 being the best. A higher F1 Score indicates a better performance. The formula for the F1 Score is mentioned in equation 7

$$\begin{aligned} \text{F1} &= \frac{2 \times (Precision \times Recall)}{Precision + Recall} \\ &= \frac{2 \times T_P}{(2 \times T_P) + F_P + F_N} \end{aligned} \tag{7}$$

6) Precision-Recall Curve: The Precision-Recall (PR) curve is a graphical evaluation metric which plots precision against recall across variable classification thresholds. PR curves are particularly used for imbalance datasets thanks to their focus on positive-class prediction quality over ROC curves. A better classification performance occurs when the PR has a larger area underneath the curve.

7) Macro Averaged Metrics: Macro averaging computes the metric independently for each class and then averages them equally, treating all classes equally regardless of the dataset size. It is represented by equation 8

$$\text{Macro Average} = \frac{1}{N}\sum_{i=1}^{N} M_i \tag{8}$$

where M_i is the metric value for each class.

8) Weighted Averaged Metrics: Weighted average deals with the average metric while taking into account the number of samples in each case and assigning varying levels of importance to data points or models, denoted by equation 9

$$\text{Weighted Average} = \frac{\sum_{i=1}^{N} w_i M_i}{\sum_{i=1}^{N} w_i} \tag{9}$$

where w_i is the class support and M_i represents the metric value for each class.

VI. RESULTS AND DISCUSSIONS

A. Classification Report Analysis

The table V presents the classification performance of the proposed MDIF framework across the three classes.

Table V
CLASSIFICATION REPORT

	Precision	Recall	F1-Score	Support
Authentic	0.92	0.92	0.92	15226
Generated	0.96	0.97	0.97	11797
Inpainted	0.68	0.66	0.67	2494
Accuracy	–	–	0.92	29517
Macro Average	0.86	0.85	0.85	29517
Weighted Average	0.92	0.92	0.92	29517

The model achieved an overall classification accuracy of 92%, showing a strong general detection capability across authentic, AI-generated and AI-manipulated images. With a precision of 0.96, a recall of 0.97 and an F1-score of 0.97, the

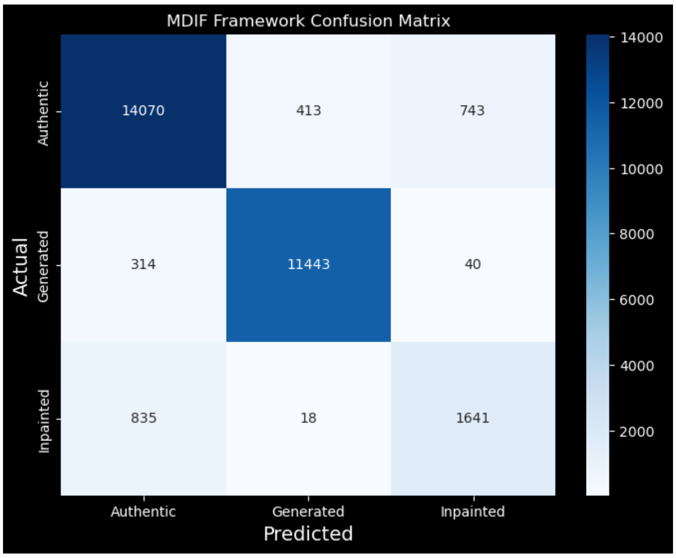

Figure 2. Confusion Matrix of the three classes: Authentic, Generated, and Inpainted

AI generated class achieved the highest performance, demonstrating the easily detectable inconsistencies these images had in their spatial and spectral domain.

The Authentic class was not too far behind with an F1 Score of 0.92, showing the model's successful rate of classification with minimal false positives and false negatives.

The Inpainted class achieved a lower performance in comparison, with a precision of 0.68, recall of 0.66 and F1score of 0.67, which was expected considering how the inpainted images contain only localized modifications with most of the image being authentic. Detection of subtle manipulations are more difficult, however, despite the challenge, the model achieved a meaningful detection capability due to the inclusion of spectral inconsistency analysis, boundary artifact detection and depth shading discrepancy analysis.

The macro average F1 Score of 0.85 is a balanced score on the performance across all the classes. The weighted average F1 score on the other hand reached a staggering 0.92 reflecting the model's stability despite the class imbalance.

B. Confusion Matrix

The confusion matrix as in figure 2 clearly illustrates the detailed prediction behavior of the model, which correctly classified 14,070 authentic images, 11,443 generated images and 1,641 inpainted images. Most of the classification errors occurred between Authentic and Inpainted classes due to the inpainted image's preservation of most of its original authentic structure, making the slight manipulation difficult to identify. The Generated class displayed least confusion with other categories during identification, confirming the strong detectable inconsistencies in these images. The confusion matrix confirms the role of the proposed hybrid feature fusion strategy in separating the three classes precisely while maintaining a reasonable detection capability for localized manipulations.

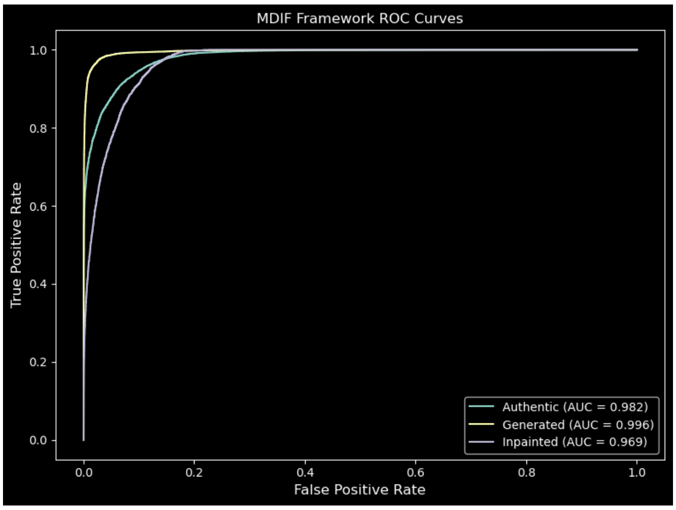

Figure 3. ROC Curves for the three classes: Authentic, Generated, and Inpainted

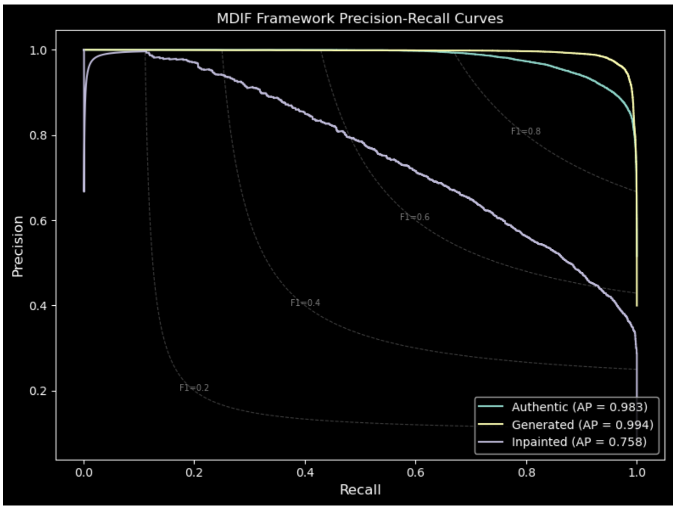

Figure 4. Precision-Recall Curves for the three classes: Authentic, Generated, and Inpainted

C. ROC Curve Analysis

The ROC curve displayed in figure 3 shows the results of the model's clear ability to distinguish the classes in the framework with the AUC values of 0.982, 0.996 and 0.969 for Authentic, Generated and Inpainted classes respectively. The Generated class's near-perfect discrimination is all because of the strong spectral and texture inconsistencies displayed by AI-generated images. Although the Inpainted class achieved a slightly slower performance, its AUC value indicates a very strong detection capability for the localized manipulations.

D. Precision-Recall Curve Analysis

The Precision-Recall curves as seen in figure 4 further validate the model's efficiency under class imbalance conditions with the Average Precision (AP) scores touching 0.983 for Authentic, 0.994 for Generated and 0.758 for Inpainted classes. The high AP scores for Authentic and Generated classes are a testimony to their stable precision across varying recall thresholds. The difficulty in detection of local manipulation is precisely why the Inpainted class achieved a lower AP score, although it is acceptable considering the complexity of Inpainting detection. These results validate the combination strategy applied in the model with the spatial, spectral and depth domain inconsistencies' tendency to improve model durability against synthetic and partially synthetic images.

VII. Conclusion

This project successfully proposed the Multi-Domain Inconsistency Fusion (MDIF) framework for distinguishing AI-generated and AI-manipulated images from authentic images using a lightweight hybrid architecture. Combining spatial RGB features, frequency-domain spectral artifacts and depth-shading geometric inconsistencies into a unified feature representation for image forgery detection, led to a display of strong overall performance reaching a 92% classification accuracy and a 0.98 AUC score. The framework was capable of exceptional performance on AI-generated images while maintaining a meaningful performance on inpainted manipulations, despite its difficulty. The results show an obvious improvement in generalisation and robustness thanks to the multi-domain fusion strategy, unlike what relying on a single feature domain would have achieved. Furthermore, the computational efficiency and ease of deployment in real-world forensic applications, even in resource-constrained environments, makes this project currently significant and a necessity. A future scope of this project would be inculcating a larger domain of inpainted images in dataset to ensure a better accuracy in that case and also test the project under compression conditions, while extending cross-generator generalisation.

VIII. Code Availability

The code used to generate the results in this paper is available at https://github.com/joejo-joestar/MDIF and a live demo of the framework can be accessed at https://mdif-app.streamlit.app.

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